

Search engine With tools agents

```
#installing libraries
```

```
!pip install langchain-google-genai
!pip install langchain-community
!pip install yfinance
```

[Show hidden output](#)

```
# import libraries
import warnings
warnings.filterwarnings("ignore")
from langchain_google_genai import ChatGoogleGenerativeAI
from pydantic import BaseModel, Field
from langchain.agents import create_agent
from langchain_community.tools import YahooFinanceNewsTool

from dotenv import load_dotenv
load_dotenv()
from google.colab import userdata
import os
```

```
from langchain_core.prompts import ChatPromptTemplate
```

```
!pip install langgraph
```

[Show hidden output](#)

```
from langgraph.checkpoint.memory import InMemorySaver
from langchain.agents.middleware import SummarizationMiddleware
```

```
# Structured output
class FinanceNews(BaseModel):
    summary:str = Field(description="Summary of the latest financial news")
    key_points:list[str]
    sentiment:str
    ticker:str

# Model
model = ChatGoogleGenerativeAI(model='gemini-1.5-flash',api_key=userdata.get("GEMINI"),temperature=0)

#Yahoo finance tool

news_tool = YahooFinanceNewsTool(top_k=5,name="yahoo_finance_news_research")

system_prompt = ChatPromptTemplate.from_messages([
    ("system",""" You are financial news reseacrh agent.
    Use the Yahoo finance news tool for the current company news.
    Summarize the important information concisely.
    Do not invent financial information """),
    ("user",{question})
])

#agent

agent = create_agent(
    tools=[news_tool],
    model=model,
    response_format=FinanceNews,
    checkpointer=InMemorySaver(),
    middleware=[
        SummarizationMiddleware(
            model=model,
            trigger=("messages",10),
            keep=("messages",4)
        )
    ]
)
```

```
#streaming
for chunk in agent.stream(
    {
        "messages":[
            {
                "role":"user",
                "content": "Find the latest news about Infosys. use the ticker INFY."
            }
        ]
    },
    config={"configurable": {"thread_id": "1"}},
    stream_mode="updates"
):
    print(chunk)
```

```
#Clean result
result = agent.invoke(
    {
        "messages": [
            {
                "role": "user",
                "content": "Find the latest news about infosys. Use ticker INFY."
            }
        ]
    },
    config={"configurable": {"thread_id": "1"}}
)

print(result["structured_response"])
```

```
#Batch processing
queries = [
    {"messages":[{"role":"user","content":"Find the latest news about Infosys. use the ticker INFY."}]},
    {"messages":[{"role":"user","content":"Find the latest news about IBM. Use ticker IBM."}]},
    {"messages":[{"role":"user","content":"Find the latest news about Tesla. Use ticker TSLA."}]},
    {"messages":[{"role":"user","content":"Find the latest news about Microsoft. Use ticker MSFT."}]},
]

results = agent.batch(queries)

for result in results:
    print(result["structured_response"])
```